Auditing Score Exposure and Presentation Order in LLM Harness Selection: A Matched 30-Task Case Study

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Abstract

Harness evaluations often choose among candidate outputs and then report a score for the chosen candidate. The interface used to make that choice is part of the measurement: a selector may see the grader's score, one of several presentation orders, or both. We conducted a within-task 2×2 audit on a frozen local benchmark, with three fixed candidate answers per task and four selection arms that crossed two orders with hidden or visible rubric scores. The audit used 30 tasks, a llama3.2:3b candidate generator, and a qwen2.5:7b selector.

Exposing scores while holding order A fixed changed the selected candidate on 7/30 tasks (23.3%; Wilson 95% CI 11.8% to 40.9%). The corresponding result under order B was 5/30 (16.7%; CI 7.3% to 33.6%). Changing order with scores hidden changed the selected slot on 15/30 tasks (50.0%; CI 33.2% to 66.8%); changing order with scores visible produced 14/30 (46.7%; CI 30.2% to 63.9%). Every changed score-exposure choice selected different answer text. Across the two order contrasts, the counts of slot changes that selected byte-identical text were 1 with scores hidden and 0 with scores visible.

The deterministic rubric most often could not distinguish the selected outputs: all three candidates tied on 24/30 tasks. Score exposure nevertheless changed the selected slot on 5/24 all-tied tasks under order A and 3/24 under order B (Table 3). When scores were visible, the selector chose a top-scored candidate in 60/60 arm-task slots. These results are a reproducible interface audit, not evidence that the rubric measures answer quality or that the observed rates generalise to other models and task populations.

1 Introduction

Evaluating an LLM agent harness usually involves more than running one fixed program. A pipeline may generate several candidates, choose one, and score the choice. Selection bias when a selected candidate is subsequently evaluated on the same evidence is a classical concern (Cawley and Talbot, 2010), and adaptive reuse of evaluation data can break the nominal validity of intervals and tests (Dwork et al., 2015). LLM evaluation adds another interface: the chooser can be an LLM that is sensitive to the way candidates are presented.

Two manipulations are easy to overlook because both leave the candidate answers unchanged. First, the selector can be shown the grader's score. If that score is also the reported outcome, the selection and measurement stages are no longer independent. Second, candidate order can change. Position effects in LLM judges are well documented (Shi et al., 2024), so an uncontrolled order difference can move choices even when no score is exposed.

We ask a narrow question. For a fixed candidate pool, how often does each manipulation change the selected slot, the selected answer text, and the selected candidate's rubric score? We answer it with one matched case study rather than a population estimate. The same three candidates enter a 2×2 design: order A or B, crossed with hidden or visible rubric scores. This lets us form two

score-exposure contrasts and two order contrasts without comparing the diagonal in which both factors change.

The contribution is an auditable empirical record:

- the exact candidate pool is shared across all four arms;
- presentation order is reconstructed from the frozen runner and checked against the analysis;
- every one of the 90 raw answer files is hashed, and every stored rubric score is recomputed;
- the paper reports slot changes, answer-content changes, and rubric-score changes separately;
- the score-exposure and order contrasts are presented under both levels of the other factor.

We do not claim a new estimator or a universal harness bias. The useful evidence is that a fixed candidate pool can produce materially different selections under two interface variations, while the benchmark's deterministic rubric often registers no score difference at all.

2 Related Work

Selection and evaluation reuse. The winner's-curse problem in model selection arises when the noisy measurement used to choose a candidate is also treated as an unbiased evaluation of that candidate (Cawley and Talbot, 2010). Reusable-holdout methods address a related problem in adaptive data analysis: repeated choices made using the same evidence can overfit that evidence (Dwork et al., 2015). Our design separates the two visible input factors before selection, but it does not provide the fresh-task holdout that would validate a deployment claim.

Presentation and preference leakage. Studies of LLM-as-a-judge systems identify position, verbosity, and self-enhancement effects (Gu et al., 2025; Shi et al., 2024). Preference leakage further shows that the relation between the generator and judge can affect judge outputs (Li et al., 2025). Work on strict estimator separation motivates care when the same information or model family participates in both choosing and evaluating (Nahhas et al., 2026). Our score-visible arm is a direct, deliberately visible interface leak rather than an inferred preference relation. The order contrast is included as a separate manipulation.

Harness evaluation. Recent agent-evaluation work treats the surrounding harness as a hidden variable rather than a constant. Harness search and recursive harness construction make the harness an object of optimisation (Xu et al., 2026); other work argues that agent comparisons must disclose the harness (Zhang et al., 2026), studies scaffold choice as part of the effective system (Vats and Golev, 2026), and examines how harness evolution should be evaluated (Wang et al., 2026). Our audit addresses a lower-level interface question within such an evaluation: what happens when the same candidates are chosen under different score visibility and order.

Estimands and construct validity. Evaluation results are difficult to interpret when the target quantity and the measurement procedure are not separated (Binette and Reiter, 2024; Freiesleben and Zezulka, 2025). We therefore report the estimator explicitly: the fraction of paired tasks on which the selected slot changes, the fraction on which selected answer content changes, and the paired difference in the selected candidate's rubric score. The rubric is an outcome measure, not an independent estimate of answer quality.

3 Audit Design

3.1 Frozen task and candidate pool

We used the development subset of a local WSE-Bench manifest. The frozen run sampled 30 tasks with seed 20261023. For each task, the generator produced 3 candidate answers from preset harness variants and wrote them to persistent files. The candidate generator was the local Ollama alias `llama3.2:3b`; the selector was the local alias `qwen2.5:7b`. Candidate generation used a low fixed temperature, and selection used deterministic decoding. The post-run installed model records are retained as artifacts; the runner itself did not persist the response's model digest.

The same three raw answers enter every selection arm. Before analysis, we verified that all 30 $\times$ 3 expected answer files existed. This prevents an arm from silently sampling a fresh candidate after the intervention.

3.2 The 2×2 selection interface

Table 1 shows the four arms. Order A and order B are deterministic, mode-specific permutations of the candidate indices. They are stable across reruns but differ on 26/30 tasks. Only within-row and within-column pairs are used. The diagonal that changes both order and score visibility is excluded.

A code audit found that the original runner seeded the visible-score order-B arm from the literal mode name rather than from the score-hidden order-B arm. The two permutations differed on 23 of 30 tasks. We therefore preserved the original run and replayed only that arm with the frozen candidates, scores, selector settings, and intended order B. The repair changed 11 selected candidates; all manuscript outcomes use the replayed record. The other three arms were reused byte-for-byte, and each replay response is stored with its prompt and response hashes.

	Scores hidden	**Scores visible**
Order A	`honest`	`hard_a`
Order B	`honest_shuffled`	`hard`

Table 1: The observed 2×2 selection design. Score-exposure contrasts compare left and right; order contrasts compare top and bottom. The one order pair is a control in this run, not a balanced sample of all possible permutations.

3.3 Outcomes

For each paired contrast, we report:

1. whether the selected candidate index changed;
2. whether the selected raw answer's SHA-256 changed, which avoids treating duplicate text as a content change;
3. the deterministic rubric score of the selected candidate before and after the manipulation.

Choice-change proportions use a Wilson score interval. Mean selected-score differences use a paired-task percentile bootstrap with 20000 draws and seed 20261027. These intervals describe uncertainty over the frozen task sample; they do not turn the case study into a population estimate. Candidate content was generated once per task and reused unchanged.

3.4 Integrity checks

The analysis and evidence are generated from the frozen runner output, not copied manually into the paper. We independently recomputed all 90 rubric scores from the raw answers with the evaluator code and obtained zero mismatches. The source JSON hash is:

`6a5275a85358c7e2e42fc16db1fdd746f7ba8e4d249e8e91b4fa26b18f530b55`

The deterministic analysis artifact hash is:

`7970e74de0a5868bc3d608fb07daf2fcec333bc019b095530e42371730ee9a4a`

The order-contract check, candidate manifest, score recomputation, and generated manuscript values are separate files in the evidence directory.

4 Results

Table 2 reports the four paired contrasts. Figure 1 shows the same choice-change proportions and rubric-score deltas.

Contrast	Slot changed	Answer changed	Rubric delta [95% CI]
Score shown, order A fixed	7/30	7	+0.0444 [+0.0000 to +0.1222]
Score shown, order B fixed	5/30	5	+0.0667 [+0.0000 to +0.1667]
Order changed, scores hidden	15/30	14	-0.0222 [-0.1000 to +0.0333]
Order changed, scores visible	14/30	14	+0.0000 [+0.0000 to +0.0000]

Table 2: Paired results. "Slot changed" uses candidate index; "answer changed" compares SHA-256 hashes of the selected raw answers. Rubric delta is the selected candidate's score under the second arm minus the first. Intervals are descriptive and condition-specific.

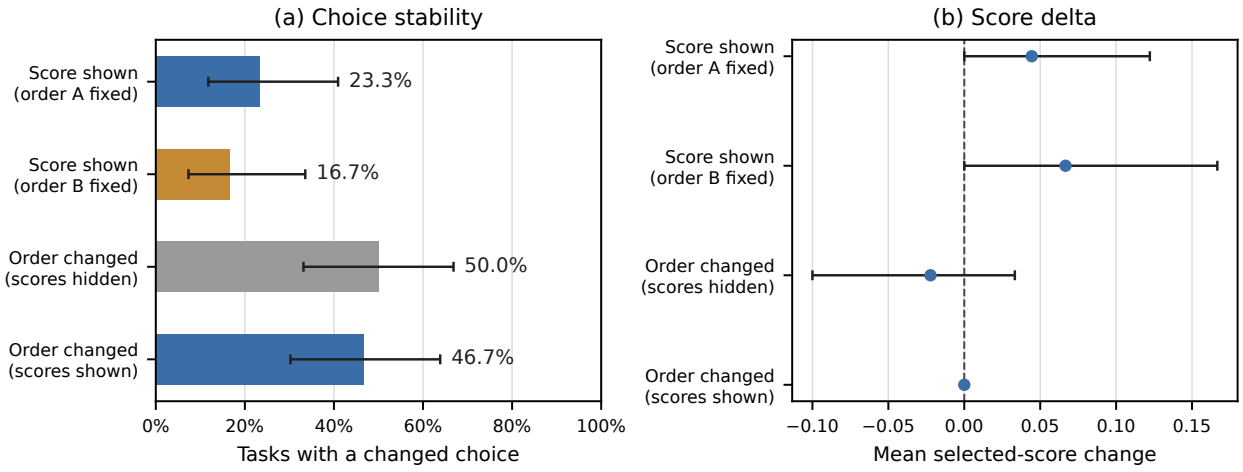


Figure 1: Matched interface contrasts. Panel (a) gives the fraction of tasks with a changed selected slot and Wilson 95% intervals. Panel (b) gives the mean selected-rubric-score change with paired-task bootstrap intervals. The four contrasts come from one order pair.

4.1 Score exposure changes choices under both orders

With order A fixed, score exposure changed the selected candidate on 7 of 30 tasks and changed selected answer content on the same number of tasks. Under order B, both counts were 5 of 30. All changed score-exposure choices therefore represented different answer text. The observed proportions, 23.3% and 16.7%, differ descriptively across the two orders, but the overlapping intervals and single order pair do not support a tested interaction.

The rubric-score changes were sparse: order A had 2 increases, 28 ties, and 0 decreases; order B had 2 increases, 28 ties, and 0 decreases. The mean changes were +0.0444 (CI +0.0000 to +0.1222) and +0.0667 (CI +0.0000 to +0.1667). These are changes in the selected candidate's rubric score, not evidence that the selected answer is better in an independent sense.

4.2 Order changes choices even without scores

When both arms withheld scores, changing order changed the selected slot on 15 of 30 tasks and selected answer content on 14. The mean rubric-score change was -0.0222 (CI -0.1000 to +0.0333), with 1 increase, 28 ties, and 1 decrease. One slot change selected byte-identical content.

When both arms showed scores, order changed the selected slot on 14 of 30 tasks and answer content on 14. None of the 14 visible-score order changes selected byte-identical content. The selected rubric score was unchanged in every paired task (30/30), so score-visible reordering moved among equally scored choices in this sample.

4.3 Ties explain why a rubric delta can miss a choice change

The coarse outcome was already visible in the candidate pool: 24 of 30 tasks had the same deterministic rubric score for all three candidates. Table 3 stratifies the score-exposure contrasts. Score exposure changed the selected slot in 5 of 24 all-tied tasks under order A, and in 3 of 24 under order B.

Score-exposure contrast	All candidates tied	Nontrivial score variation	Total
Order A fixed	5/24	2/6	7/30
Order B fixed	3/24	2/6	5/30

Table 3: Slot changes after exposing scores, stratified by whether the three frozen rubric scores were all equal. The all-tied stratum has no numerical reason to prefer one candidate under the published score, yet the selector sometimes moved between candidates.

The score-visible instrument check was active: both score-visible arms chose a top-scored candidate in 60/60 arm-task slots, including 12/12 slots with nontrivial score variation. The all-tied selections account for many of the remaining slots because every candidate is top-scored by definition. This check establishes that the displayed scores reached the selector; it does not identify a cognitive mechanism for the all-tied changes.

5 Discussion

Selection is part of the reported evaluation. The candidate answers were fixed, but the selected candidate index changed after score visibility or order changed. A pipeline that reports only the chosen candidate's score can therefore hide an interface dependence that remains visible

when the full choice record is retained. The appropriate unit of analysis includes the candidate pool, the selector, and the presentation interface.

Choice changes and score changes are different outcomes. Score exposure changed choices under both orders, but the deterministic rubric changed on few tasks. That pattern follows from the large all-tied stratum: moving from one candidate to another need not change the rubric. Reporting only a mean score would miss most of the observed interface sensitivity, while reporting only choice instability would overstate what is known about answer quality.

Order requires an explicit control. The order-only contrast changed slots and answer content while scores remained hidden. This is consistent with the broader position-bias literature (Shi et al., 2024); the present result adds a matched record for this selector and task sample. It also shows why score exposure must be evaluated at a fixed order. Comparing one hidden-order arm with a score-visible arm in a different order would combine both factors.

The rubric is an outcome, not a referee. This benchmark's deterministic evaluator records whether required content appears in a response. It can make many candidates tie and cannot establish whether a response's reasoning is sound. The paper therefore does not convert a rubric delta into a quality claim, and a zero mean delta does not show that the interface manipulation was harmless.

6 Limitations

1. **One model pair and one order pair.** The generator and selector aliases are fixed, and only two candidate permutations are observed. The study does not estimate an average over models or all permutations.
2. **One deterministic selection per cell.** The design measures sensitivity to the prompt interface in this configuration; it does not estimate run-to-run variance of a stochastic selector.
3. **A coarse local benchmark.** Most tasks have an all-tied candidate pool, and the rubric is not independent ground truth. The sample is a development subset, not a fresh or hidden evaluation.
4. **No population inference.** Wilson and bootstrap intervals describe uncertainty over this frozen task sample. They do not repair the single-model, single-order design.
5. **Model aliases were not logged per response.** Post-run local records identify the installed model digests, but the runner did not persist a digest with every call.
6. **No claim about upstream harness benefit.** The experiment evaluates how a selector responds to interface factors. It does not establish that any harness variant improves task performance.

7 Conclusion

On a frozen 30-task candidate pool, both score visibility and candidate order changed the selector's chosen candidate. Score exposure changed selected answer content on 7 tasks under order A and 5

under order B. Order changed the selected slot on 15 tasks without scores and 14 with scores. Slot changes selecting byte-identical content numbered 1 without scores and 0 with scores.

The rubric often could not distinguish the candidates: 24 of 30 tasks had an all-tied pool. This makes choice-level auditing necessary even when a score-level comparison looks stable. The result is local and descriptive; its practical value is the separated, hash-checked record and the demonstration that score exposure and order should be reported as distinct interface factors.

Data and Code Availability

This draft is accompanied by a local reproducibility directory containing the frozen source record, raw candidate manifest, independent score recomputation, order-contract check, deterministic analysis, generated figure, and verification scripts. Appendix A records the reproduction checks. No public repository or DOI is claimed in this draft. The source run is identified by SHA-256:

`6a5275a85358c7e2e42fc16db1fdd746f7ba8e4d249e8e91b4fa26b18f530b55`

The recomputed analysis is identified by:

`7970e74de0a5868bc3d608fb07daf2fcec333bc019b095530e42371730ee9a4a`

Appendix B records the legacy claims excluded from this revision.

A Reproduction Record

The experiment command, model aliases, source record, and environment observations were captured alongside the manuscript. The analysis can be regenerated from the source JSON without calling a model. The following checks are separate evidence files:

- raw candidate manifest: 90 expected files, 90 captured, zero missing;
- order repair: 23/30 original order-B mismatches identified, only the affected arm replayed, and 11 picks changed;
- score recomputation: 90 stored scores reproduced exactly, zero mismatches;
- order contract: the factorial mapping is present and the two orders differ on 26/30 tasks;
- content comparison: selected raw answers were hashed before the slot-change counts were interpreted;
- manuscript values: generated from the analysis JSON and the captured run configuration.

The deterministic analysis was run twice and produced byte-identical JSON. The original manuscript and all earlier revision directories were preserved.

B Claims Excluded from This Revision

Earlier drafts contained a judge-inflation comparison, Cohen's κ as an effect measure, and a simulation that treated selection benefit as bias. Those analyses are excluded here. The judge-inflation records did not consistently bind judged answers to their declared source files; the kappa comparison did not have an effect-test null hypothesis; and the simulation's estimand mixed selected-object

quality with cross-task strategy value. None of those legacy values is used in the abstract, results, figures, or conclusion of this manuscript.

This revision also omits the earlier benchmark-defect extrapolation and any claim that anonymity removes presentation effects. The present result is narrower: for one fixed candidate pool, two selection-interface factors changed choices, while the deterministic rubric often did not register a score change.

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